ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) | 979-8-3503-6874-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICASSP49660.2025.10890826

Dual Position Attention Time-Frequency Network for Binaural Audio Synthesis

Changjun He
Harbin Institute of Technology
Harbin, China
changjunhe@stu.hit.edu.cn

Weiping Chen*
Harbin Institute of Technology
Harbin, China
chenwp@hit.edu.cn

Mingjiang Wang
Harbin Institute of Technology (Shenzhen)
Shenzhen, China
mjwang@hit.edu.cn

***Abstract*—In applications such as virtual reality and augmented reality, binaural audio provides listeners with a more immersive experience. To synthesize binaural audio with enhanced spatial localization, especially in scenarios involving moving sound sources, accurate phase estimation is crucial. However, existing deep learning methods have yet to achieve satisfactory results in this area. To address this issue, this paper introduces a Dual Position Attention Time-Frequency Network (DPATFNet). Specifically, our approach targets the interaural differences and Doppler effects induced by sound source movement, guiding the synthesis process from monaural to binaural audio through strong positional conditions in the time-frequency domain. The network employs a Dual Position Attention Block (DPAB) to effectively focus on sound source movement and improve phase estimation performance. The proposed DPATFNet demonstrates a strong capability for synthesizing accurate binaural audio, with experimental results on the Binaural Speech dataset showing that DPATFNet achieves state-of-the-art performance in phase metrics (Phase-L2: 0.717, IPD-L2: 1.020, Wave-L2: 0.148, Amplitude-L2: 0.037).**

***Index Terms*—Binaural audio synthesis, dual position attention, phase estimation, time-frequency analysis**

I. Introduction

Spatial audio is a distinct format from conventional mono and stereo formats, offering listeners an immersive experience, often referred to as 3D audio [1]. The reproduction and playback of spatial audio involve multichannel speaker system rendering and binaural rendering [2], [3]. Among these, binaural rendering is particularly prevalent in virtual reality and augmented reality applications due to its reduced dependency on hardware systems. This technology enhances auditory realism and spatial perception, thereby compensating for the visual limitations inherent in virtual meetings, gaming, and online concerts [4]–[6].

Binaural audio synthesis is initially implemented using traditional Digital Signal Processing (DSP) methods, which model the propagation of sound from the source to the human ear as a linear time-invariant system. This approach simulates various components of the sound at the listener's ears, including direct sound, early reflections, late reverberations, and Head-Related Impulse Responses (HRIR) [7]–[11]. While DSP-based methods are straightforward and interpretable, they fail to effectively model the nonlinearities present in sound propagation, leading to suboptimal binaural audio quality [12]. Consequently, with the increasing maturity of deep learning techniques in audio processing [13]–[15], end-to-end audio binauralization methods based on deep neural networks are proposed [16]. These methods generally fall into two categories: time-domain and frequency-domain. Time-domain waveform-based methods are shown to produce more accurate binaural audio, leading to the development of numerous time-domain binauralization networks that demonstrate promising results [17], [18]. In contrast, frequency-domain approaches, although less accurate, exhibit stronger generalization capabilities, particularly for the audio from the out-of-distribution domain [19]. Additionally, some studies employ deep neural networks to directly predict Head-Related Transfer Functions (HRTFs) and impulse responses [20]–[22]. Multimodal methods that incorporate video information have also been extensively studied for binaural audio synthesis [23]–[31]. However, in these studies, the input is often pseudo-monaural audio derived from the mixing of left and right ear recordings, which cannot truly simulate the propagation process from the sound source to the human ear. Moreover, existing methods still struggle with accurate phase estimation in binaural audio, which is essential for spatial perception, because the human ear can detect Interaural Time Differences (ITD) as small as 100 μs [16], [22], [32], and sound sources are primarily located through the Interaural Phase Difference (IPD) between the left and right ears.

In this paper, we focus on the process of sound propagation and introduce a Dual Position Attention Time-Frequency Network (DPATFNet). Under strong positional conditions, DPATFNet initially warps the monaural audio in the time domain, followed by multiple upsampling steps applied to the time-frequency features of the initially warped binaural audio, ultimately generating the final binaural output. A Dual Position Attention Block (DPAB) is specifically designed to integrate positional information and capture the Doppler effect induced by sound source movements, thereby further guiding the binaural audio synthesis in the frequency domain. The contributions of this work can be summarized as follows:

1. The proposed DPATFNet simultaneously focuses on both time and frequency domains, effectively modeling interaural

Thanks to the National Natural Science Foundation of China under Grant No.62276076 and No.62176102 and the Guangdong Basic and Applied Basic Research Foundation under Grant No. 2023A1515140109.

∗ Corresponding author

Authorized licensed use limited to: National Yang Ming Chiao Tung University. Downloaded on September 09,2025 at 08:29:11 UTC from IEEE Xplore. Restrictions apply.

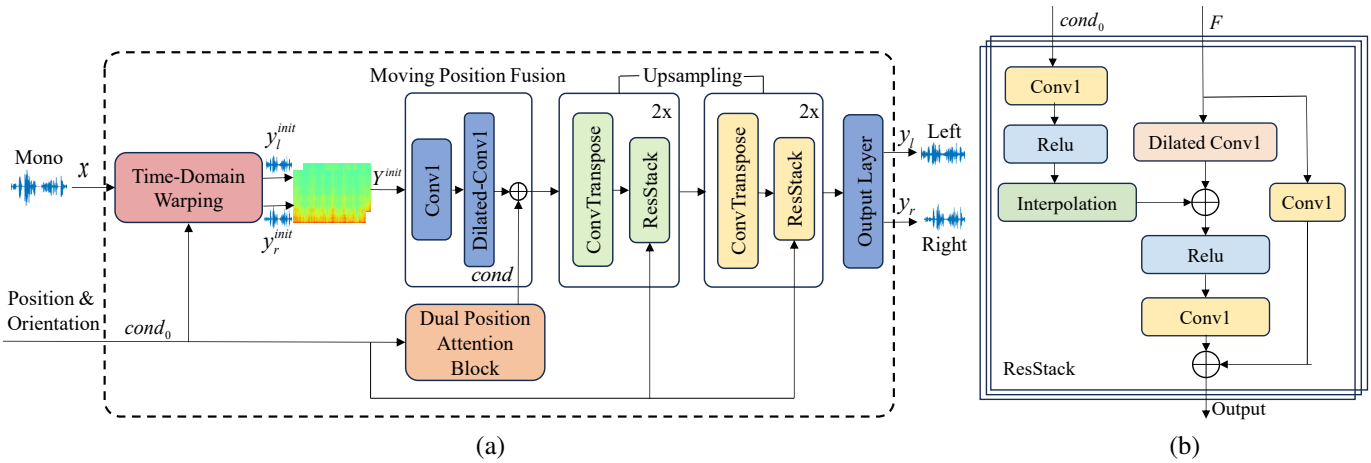

Fig. 1. (a) The overall structure of the proposed Dual Position Attention Time-Frequency Network (DPATFNet). (b) The architecture of ResStack.

differences in binaural audio.

2. We introduce a dual position attention block that effectively simulates the Doppler effect, which is crucial for addressing sound source movements in binaural audio synthesis tasks.

3. Experimental results on the Binaural Speech dataset demonstrate that DPATFNet can accurately synthesize binaural audio, achieving state-of-the-art performance in phase metrics, and it excels in the binauralization of audio from out-of-distribution domains.

II. Proposed Method

A. Overall Structure

The overall structure of the DPATFNet is illustrated in Fig. 1(a). The primary objective of this model is to synthesize binaural audio $y = (y_l, y_r) \in \mathbb{R}^{B\times 2\times L}$ from a given monaural audio input $x \in \mathbb{R}^{B\times 1\times L}$ by incorporating the position and orientation of the sound source as an auxiliary condition $cond_0 \in \mathbb{R}^{B\times 7\times T}$. Here, T denotes the length of the conditional time series, L denotes the audio length within T, and the sampling rate of the conditional information is much lower than the audio sampling rate. The dimension 7 denotes that each record of conditional information includes 7 channels: 3 for positional information and 4 for orientation information. Our model first follows the design in [16] and utilizes a Time-Domain Warping (TDW) module to warp the monaural audio x in the time domain, thereby simulating ITD caused by the sound propagation from the source to the ears.

After the initial time warping, the preliminary binaural audio $y^{init} = (y_l^{init}, y_r^{init}) \in \mathbb{R}^{B\times 2\times L}$ is obtained. To achieve more accurate modeling of IPD and other spatial cues, further processing is performed in the frequency domain. Specifically, the warped audio y^{init} is first transformed using Short-Time Fourier Transform (STFT), resulting in time-frequency features (Y_l^{init}, Y_r^{init}), which are concatenated along the frequency dimension to form the combined time-frequency feature Y^{init}. Next, to effectively capture the Doppler effect and interaural phase differences caused by sound source movement in frequency domain [33], [34], the output $cond$ from DPAB is incorporated into the shallow time-frequency features via Moving Position Fusion (MPF) layer. The fused time-frequency features are then upsampled using multiple transposed convolution layers, following the design in [35], with dilated convolutional residual blocks added after each transposed convolution layer. To ensure that the binauralization process is strongly guided by position and orientation, $cond_0$ is added as an auxiliary condition to each ResStack block. After upsampling, the synthesized binaural audio $y = (y_l, y_r)$ is produced by Conv1 and Tanh output layers. The model's overall process can be represented as follows:

$$(y_l, y_r) = \text{Tanh}(\text{Conv1}(f_1(\text{STFT}(f_2), f_3, cond_0))) \quad (1)$$

where f_1 denotes the MPF and Upsampling process, f_2 denotes the TDW module, and f_3 denotes the DPAB module, as described by the following equation:

$$\begin{aligned} f_2 &= \textbf{TDW}(x \mid cond_0) \rightarrow (y_l^{init}, y_r^{init}) \\ f_3 &= \textbf{DPAB}(cond_0) \rightarrow cond \end{aligned} \quad (2)$$

B. Dual Position Attention Block

In the task of synthesizing binaural audio for moving sound sources, it is crucial to consider the Doppler effect and other factors to ensure the realism of the synthesized audio. The DPAB specifically focuses on the moments when the positional changes occur. The architecture of the DPAB module is illustrated in Fig. 2(a). The first Position Attention Layer (PAL_1) takes the conditional information $cond_0$ as input and employs cross-attention to fuse the orientation with positional information of the sound source, resulting in a fused conditional feature $cond_1$. The second Position Attention Layer (PAL_2) aims to capture the precise moments of sound source movement by performing self-attention [36] on the output $cond_1$ from PAL_1, extracting the information related to positional shifts, which is denoted as $cond_2$. Finally, $cond_2$ is convolved and then concatenated with $cond_0$ to

Authorized licensed use limited to: National Yang Ming Chiao Tung University. Downloaded on September 09,2025 at 08:29:11 UTC from IEEE Xplore. Restrictions apply.

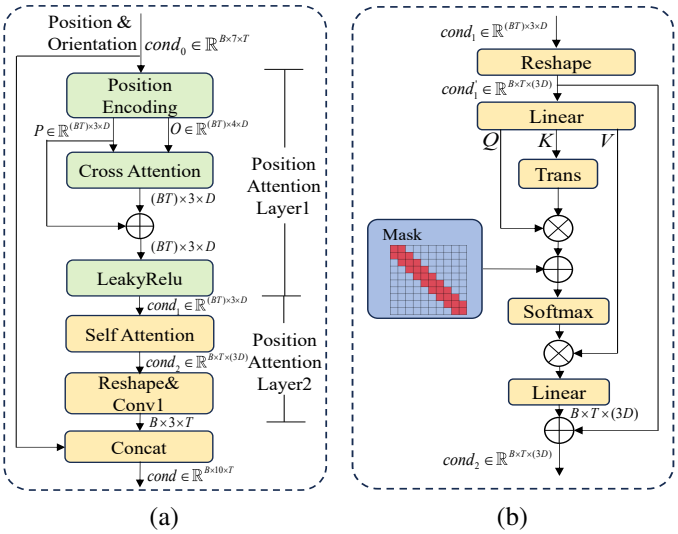

Fig. 2. (a) The architecture of Dual Position Attention Block. (b) The architecture of the second Position Attention Layer.

form the final conditional information $cond$. This process can be described as follows, where Atten_1 and Atten_2 denotes the functions of the PAL_1 and PAL_2, respectively. Concat denotes the concatenation of the convolved $cond_2$ with the input conditional information $cond_0$.

$$cond = \mathrm{Concat}(\mathrm{Atten}_2(\mathrm{Atten}_1(cond_0)), cond_0) \tag{3}$$

For PAL_1, we follow the design in [19]. Initially, sinusoidal encoding is applied to obtain the encoded positional information $P \in \mathbb{R}^{(BT)\times 3\times D}$ and orientation information $O \in \mathbb{R}^{(BT)\times 4\times D}$, where D denotes the dimension of each channel. The encoded positional information P is then used as the query, while the encoded orientation information O serves as the key and value, enabling cross-attention fusion. Finally, P is added to the cross-attention output through a residual connection, resulting in the fused condition feature $cond_1 \in \mathbb{R}^{(BT)\times 3\times D}$.

The architecture of PAL_2 is shown in Fig. 2(b). First, the fused condition feature $cond_1$ is reshaped into $cond_1^{'} \in \mathbb{R}^{B\times T\times (3D)}$. Then the conditional feature $cond_1^{'}$ is passed through three linear layers to generate the query ($Q \in \mathbb{R}^{B\times T\times (3D)}$), key ($K \in \mathbb{R}^{B\times T\times (3D)}$), and value ($V \in \mathbb{R}^{B\times T\times (3D)}$), respectively, followed by the computation of attention weights. Since we only need to focus on the moments when positional changes occur, a mask is applied to the time dimension. This mask ensures that, for each moment t, attention is directed only to the conditional information at $t-1$ and $t+1$, ignoring relationships with conditional information at more distant time frames. Subsequently, the attention weight matrix with the applied mask is normalized using softmax and multiplied by V. Finally, the conditional feature $cond_1^{'}$ is added to the attention output through a residual connection, yielding $cond_2$, which encapsulates the information about positional change.

C. ResStack with Condition Injection

For the ResStack, we follow the design in [35], where each ResStack contains three dilated convolution layers, as illustrated in Fig. 1(b). The input audio feature for the current ResStack is represented as $F \in \mathbb{R}^{B\times C\times W}$, where C denotes the number of channels, and W denotes the width of the current feature. To integrate the conditional information $cond_0$ into this structure, we design a conditional injector. First, $cond_0$ is passed through a Conv1 and ReLU layer to match its channel dimension with that of the current ResStack. Additionally, since the sampling rate of the conditional information is much lower than that of the audio, interpolation is applied to align its size with the current ResStack. The interpolated conditional information is then added as an auxiliary condition after each dilated convolution layer. Finally, the main pathway and the shortcut pathway are each processed through a Conv1 layer and then summed to produce the final output.

D. Loss Function

During training, the model is optimized by minimizing the weighted sum of the L2 loss and the phase loss between the synthesized binaural audio y and the ground truth y^*. The inclusion of the phase loss effectively compensates for the limitations of the L2 loss [16]. The loss function is defined as follows, where λ_1 and λ_2 represent the loss weights.

$$L(y, y^*) = \lambda_1 L_2(y, y^*) + \lambda_2 L_{phase}(y, y^*) \tag{4}$$

III. Experiments

A. Datasets

We use the Binaural Speech dataset published in [16] for training and testing. This dataset includes binaural audio, corresponding monaural audio, as well as orientation and positional information of the sound source and listener, collected from 8 speakers while the listener remains stationary and the speakers are in motion. The audio is recorded at a sampling rate of 48 kHz, with approximately 15 minutes of data per speaker, totaling 2 hours. The orientation and positional information are tracked and recorded at a sampling rate of 120 Hz. For consistency and ease of comparison, we follow the original split of training/valid/test sets in [16].

B. Experimental Setup

For the STFT, we use a window size of 1024 samples and a hop size of 64 samples. In the upsampling process, we employ four transposed convolution layers with upsampling factors of 4, 4, 2, and 2, respectively. The batch size is set to 32. The model parameters are optimized using the Adam optimizer [38], with the learning rate adjusted by the Cosine Annealing strategy [39], ranging from a maximum of 1×10^{-3} to a minimum of 1×10^{-6}. The loss weights λ_1 and λ_2 are set to 1.0 and 0.01, respectively.

Authorized licensed use limited to: National Yang Ming Chiao Tung University. Downloaded on September 09,2025 at 08:29:11 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF OUR APPROACH WITH ADVANCED MODELS ON THE BINAURAL SPEECH DATASET AND AUDIO FROM OUT-OF-DISTRIBUTION DOMAIN.

Model	Year	Wave-L2↓	Amplitude-L2↓	Phase-L2↓	IPD-L2↓	MOS↑	Spatialization MOS↑	Similarity MOS↑
WaveNet [37]	2016	0.188	0.035‡	0.860	1.114	3.84±0.17	3.87±0.23	3.64±0.33
WarpNet [16]	2021	0.167	0.048	0.807‡	1.166	3.88±0.15	3.93±0.23	3.84±0.26
WarpNet*,[a] [17]	2022	0.157	0.038	0.838	-	-	-	-
BinauralGrad [17]	2022	**0.128**†,[b]	**0.030**†	0.837	1.099‡	3.72±0.26	3.70±0.22	3.98±0.28
NFS [19]	2023	0.172	0.035‡	0.999	1.250	3.79±0.19	3.77±0.22	3.80±0.30
DPATFNet	2024	0.148‡,[b]	0.037	**0.717**†	**1.020**†	**3.93**±0.18	**4.05**±0.21	**4.04**±0.26

[a] WarpNet* shows the results reported in [17].
[b] The highest score is marked with † and highlighted in bold. The second highest score is marked with ‡.

C. Evaluation Metrics

Both quantitative and perceptual evaluations are conducted to evaluate the quality of the synthesized binaural audio. The quantitative evaluation includes the Mean Squared Error (MSE) between the synthesized binaural audio and the groundtruth in various aspects: 1) Wave-L2: MSE on wave; 2) Phase-L2: MSE on the phase after performing STFT; 3) Amplitude-L2: MSE on the amplitude after performing STFT; 4) IPD-L2: MSE on the IPD after performing STFT [40]. The perceptual evaluation includes: 1) Mean Option Score (MOS): the overall naturalness and fluency of the binaural audio; 2) Spatialization MOS: the perceived spatial quality of the binaural audio; 3) Similarity MOS: the perceived similarity between the synthesized binaural audio and the ground truth. For out-of-domain audio, Similarity MOS is not evaluated due to the lack of real binaural audio.

IV. RESULTS AND DISCUSSION

A. Comparison with Advanced Binaural Audio Synthesis Models

Table I presents the comparison results between the proposed DPATFNet [1] and advanced binaural audio synthesis models. In the WaveNet model, the position and orientation of the sound source are incorporated as an additional condition, applied after the dilated convolutions. In terms of quantitative evaluation, DPATFNet records the second-best Wave-L2 error and achieves the best performance in Phase-L2 and IPD-L2 error, which is outstanding because phase estimation performance is the most essential in binaural audio. Compared to the classical WarpNet model, DPATFNet demonstrates superior performance across all evaluation metrics. For perceptual evaluation, 20 participants are invited to rate each model in terms of naturalness, spatiality, and similarity, with scores ranging from 1 to 5, where higher scores indicate better performance. The MOS results with the 95% confidence interval indicate that our model outperforms other models in all aspects, with a particular advantage in spatial perception. Additionally, each model is used to binauralize music, songs, and other sounds. Our model exhibits superior performance on the audio from the out-of-distribution domain, comparable to NFS. Although BinauralGrad performs well in quantitative evaluation, particularly Wave-L2 and Amplitude-L2, it shows poor performance on the audio from the out-of-distribution domain, especially for music.

[1]Test demos are available: https://hehelita.github.io/DPATFNet/

TABLE II
ABLATION EXPERIMENT RESULTS OF DPATFNET.

Model	Wave-L2↓	Amplitude-L2↓	Phase-L2↓	IPD-L2↓
DPATFNet	**0.148**	**0.037**	**0.717**	**1.020**
w/o TDW	0.203	0.040	0.832	1.048
w/o DPAB	0.164	0.038	0.740	1.074
w/o MPF	0.162	**0.037**	0.768	1.104

B. Ablation Study

The impact of various modules within DPATFNet is assessed through ablation studies, detailed in Table II. w/o TDW, which does not model ITD in the time domain waveform, shows worse performance across all metrics, especially in Wave-L2 and Phase-L2. w/o DPAB demonstrates that the dual position attention block effectively enhances phase estimation performance, but it has no significant effect on amplitude estimation performance. w/o MPF bypasses the moving position fusion layer and instead applies the position attention block directly during the upsampling process. While this approach does not significantly impact amplitude performance, it results in worse Wave-L2 scores and phase estimation performance. This outcome is likely because the Doppler effect primarily conveys information in the frequency domain, making it more effective to incorporate positional change information into the shallow time-frequency features.

V. CONCLUSION

This paper introduces a Dual Position Attention Time-Frequency Network (DPATFNet), which models interaural differences in both the time and frequency domains. The inclusion of the Dual Position Attention Block (DPAB) effectively captures the Doppler effect during sound source movement, significantly improving the accuracy of phase estimation in the task of binaural audio synthesis. The performance of DPATFNet on the Binaural Speech dataset and audio from the out-of-distribution domain demonstrates its outstanding design. Future research directions include further enhancing the spatialization and realism of binaural audio, as well as improving the network's generalization capability to be applicable across arbitrary audio domains.

Authorized licensed use limited to: National Yang Ming Chiao Tung University. Downloaded on September 09,2025 at 08:29:11 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] F. Rumsey, *Spatial audio*. Routledge, 2012.

[2] H. Hacihabiboglu, E. De Sena, Z. Cvetkovic, J. Johnston, and J. O. Smith III, "Perceptual spatial audio recording, simulation, and rendering: An overview of spatial-audio techniques based on psychoacoustics," *IEEE Signal Processing Magazine*, vol. 34, no. 3, pp. 36–54, 2017.

[3] B. Rafaely, V. Tourbabin, E. Habets, Z. Ben-Hur, H. Lee, and H. Gamper, "Spatial audio signal processing for binaural reproduction of recorded acoustic scenes–review and challenges," *Acta Acustica*, vol. 6, p. 47, 2022.

[4] C. Hendrix and W. Barfield, "The sense of presence within auditory virtual environments," *Presence: Teleoperators & Virtual Environments*, vol. 5, no. 3, pp. 290–301, 1996.

[5] M. L. Hawley, R. Y. Litovsky, and J. F. Culling, "The benefit of binaural hearing in a cocktail party: Effect of location and type of interferer," *The Journal of the Acoustical Society of America*, vol. 115, no. 2, pp. 833–843, 2004.

[6] J. Shelton and G. P. Kumar, "Comparison between auditory and visual simple reaction times," *Neuroscience and medicine*, vol. 1, no. 01, pp. 30–32, 2010.

[7] L. Savioja, J. Huopaniemi, T. Lokki, and R. Väänänen, "Creating interactive virtual acoustic environments," *Journal of the Audio Engineering Society*, vol. 47, no. 9, pp. 675–705, 1999.

[8] K. Sunder, J. He, E. L. Tan, and W. S. Gan, "Natural sound rendering for headphones: integration of signal processing techniques," *IEEE Signal Processing Magazine*, vol. 32, no. 2, pp. 100–113, 2015.

[9] D. N. Zotkin, R. Duraiswami, and L. S. Davis, "Rendering localized spatial audio in a virtual auditory space," *IEEE Transactions on multimedia*, vol. 6, no. 4, pp. 553–564, 2004.

[10] J. W. Lee, S. Lee, and K. Lee, "Global hrtf interpolation via learned affine transformation of hyper-conditioned features," in *2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023, pp. 1–5.

[11] W. Zhang, P. N. Samarasinghe, H. Chen, and T. D. Abhayapala, "Surround by sound: A review of spatial audio recording and reproduction," *Applied Sciences*, vol. 7, no. 5, p. 532, 2017.

[12] F. Brinkmann, A. Lindau, and S. Weinzierl, "On the authenticity of individual dynamic binaural synthesis," *The Journal of the Acoustical Society of America*, vol. 142, no. 4, pp. 1784–1795, 2017.

[13] D. Michelsanti, Z. Tan, S. Zhang, Y. Xu, M. Yu, and D. Yu, "An overview of deep-learning-based audio-visual speech enhancement and separation," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 29, pp. 1368–1396, 2021.

[14] N. Li, S. Liu, Y. Liu, S. Zhao, and M. Liu, "Neural speech synthesis with transformer network," in *Proceedings of the AAAI conference on artificial intelligence (AAAI)*, vol. 33, no. 01, 2019, pp. 6706–6713.

[15] Z. Mu, X. Yang, and Y. Dong, "Review of end-to-end speech synthesis technology based on deep learning," *arXiv preprint arXiv:2104.09995*, 2021.

[16] A. Richard, D. Markovic, I. D. Gebru, S. Krenn, G. A. Butler, and F. Torre, "Neural synthesis of binaural speech from mono audio," in *International Conference on Learning Representations (ICLR)*, 2021.

[17] Y. Leng, Z. Chen, J. Guo, H. Liu, J. Chen, and X. Tan, "Binauralgrad: A two-stage conditional diffusion probabilistic model for binaural audio synthesis," *Advances in Neural Information Processing Systems*, vol. 35, pp. 23 689–23 700, 2022.

[18] W. C. Huang, D. Markovic, A. Richard, I. D. Gebru, and A. Menon, "End-to-end binaural speech synthesis," *arXiv preprint arXiv:2207.03697*, 2022.

[19] J. W. Lee and K. Lee, "Neural fourier shift for binaural speech rendering," in *2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023, pp. 1–5.

[20] I. D. Gebru, D. Marković, A. Richard, S. Krenn, G. A. Butler, and F. De la Torre, "Implicit hrtf modeling using temporal convolutional networks," in *2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2021, pp. 3385–3389.

[21] Y. Zhou, H. Jiang, and V. K. Ithapu, "On the predictability of hrtfs from ear shapes using deep networks," in *2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2021, pp. 441–445.

[22] A. Richard, P. Dodds, and V. K. Ithapu, "Deep impulse responses: Estimating and parameterizing filters with deep networks," in *2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 3209–3213.

[23] K. K. Parida, S. Srivastava, and G. Sharma, "Beyond mono to binaural: Generating binaural audio from mono audio with depth and cross modal attention," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2022, pp. 3347–3356.

[24] R. Gao and K. Grauman, "2.5 d visual sound," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 324–333.

[25] K. Yang, B. Russell, and J. Salamon, "Telling left from right: Learning spatial correspondence of sight and sound," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 9932–9941.

[26] W. Zhang and J. Shao, "Multi-attention audio-visual fusion network for audio spatialization," in *Proceedings of the 2021 International Conference on Multimedia Retrieval (ICMR)*, 2021, pp. 394–401.

[27] Y. Lu, H. Lee, H. Tseng, and M. Yang, "Self-supervised audio spatialization with correspondence classifier," in *2019 IEEE International Conference on Image Processing (ICIP)*. IEEE, 2019, pp. 3347–3351.

[28] K. K. Rachavarapu, V. Sundaresha, and A. Rajagopalan, "Localize to binauralize: Audio spatialization from visual sound source localization," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 1930–1939.

[29] Y. Lin and Y. F. Wang, "Exploiting audio-visual consistency with partial supervision for spatial audio generation," in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, vol. 35, no. 3, 2021, pp. 2056–2063.

[30] S. Li, S. Liu, and D. Manocha, "Binaural audio generation via multi-task learning," *ACM Transactions on Graphics (TOG)*, vol. 40, no. 6, pp. 1–13, 2021.

[31] Z. Li, B. Zhao, and Y. Yuan, "Cross-modal generative model for visual-guided binaural stereo generation," *Knowledge-Based Systems*, vol. 296, p. 111814, 2024.

[32] C. P. Brown and R. O. Duda, "A structural model for binaural sound synthesis," *IEEE transactions on speech and audio processing*, vol. 6, no. 5, pp. 476–488, 1998.

[33] T. P. Gill, *The Doppler effect : an introduction to the theory of the effect*. Logos Press, 1965.

[34] Y. Iwaya and Y. Suzuki, "Rendering moving sound with the doppler effect in sound space," *Applied Acoustics*, vol. 68, no. 8, pp. 916–922, 2007.

[35] K. Kumar, R. Kumar, T. De Boissiere, L. Gestin, W. Z. Teoh, and J. Sotelo, "Melgan: Generative adversarial networks for conditional waveform synthesis," *Advances in neural information processing systems*, vol. 32, 2019.

[36] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, and A. N. Gomez, "Attention is all you need," *Advances in Neural Information Processing Systems*, 2017.

[37] A. Van Den Oord, S. Dieleman, H. Zen, K. Simonyan, O. Vinyals, and A. Graves, "Wavenet: A generative model for raw audio," *arXiv preprint arXiv:1609.03499*, vol. 12, 2016.

[38] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.

[39] I. Loshchilov and F. Hutter, "Sgdr: Stochastic gradient descent with warm restarts," *arXiv preprint arXiv:1608.03983*, 2016.

[40] Y. Li, Y. Shen, and D. Wang, "Diffbas: An advanced binaural audio synthesis model focusing on binaural differences recovery," *Applied Sciences*, vol. 14, no. 8, p. 3385, 2024.

Authorized licensed use limited to: National Yang Ming Chiao Tung University. Downloaded on September 09,2025 at 08:29:11 UTC from IEEE Xplore. Restrictions apply.